Subtitle Basic idea about DC to AC converter
Institute Dr B C Roy Engineering College, Durgapur
Subject Power Electronics EE 603

Inverter Lecture 09

Kingsuk Majumdar

Hello friends

Lecture 09

1. Brief recap of last lecture 08
2. Current Source Inverter (CSI)
3. Signification
4. References

Why CSI

A CSI is a dual of VSI. In a VSI, the load current depends on the load impedance but in case of in a CSI the load voltage is depending on the load impedance. Hence, diodes are connected in series with switching devices (instead of anti-parallel, VSI) to protect them from transient voltage due to load current switching.

Where CSI

- speed control of ac motor
- Induction heating
- lagging VAR compensation
- Sychronous motor starting

CSI basic circuit

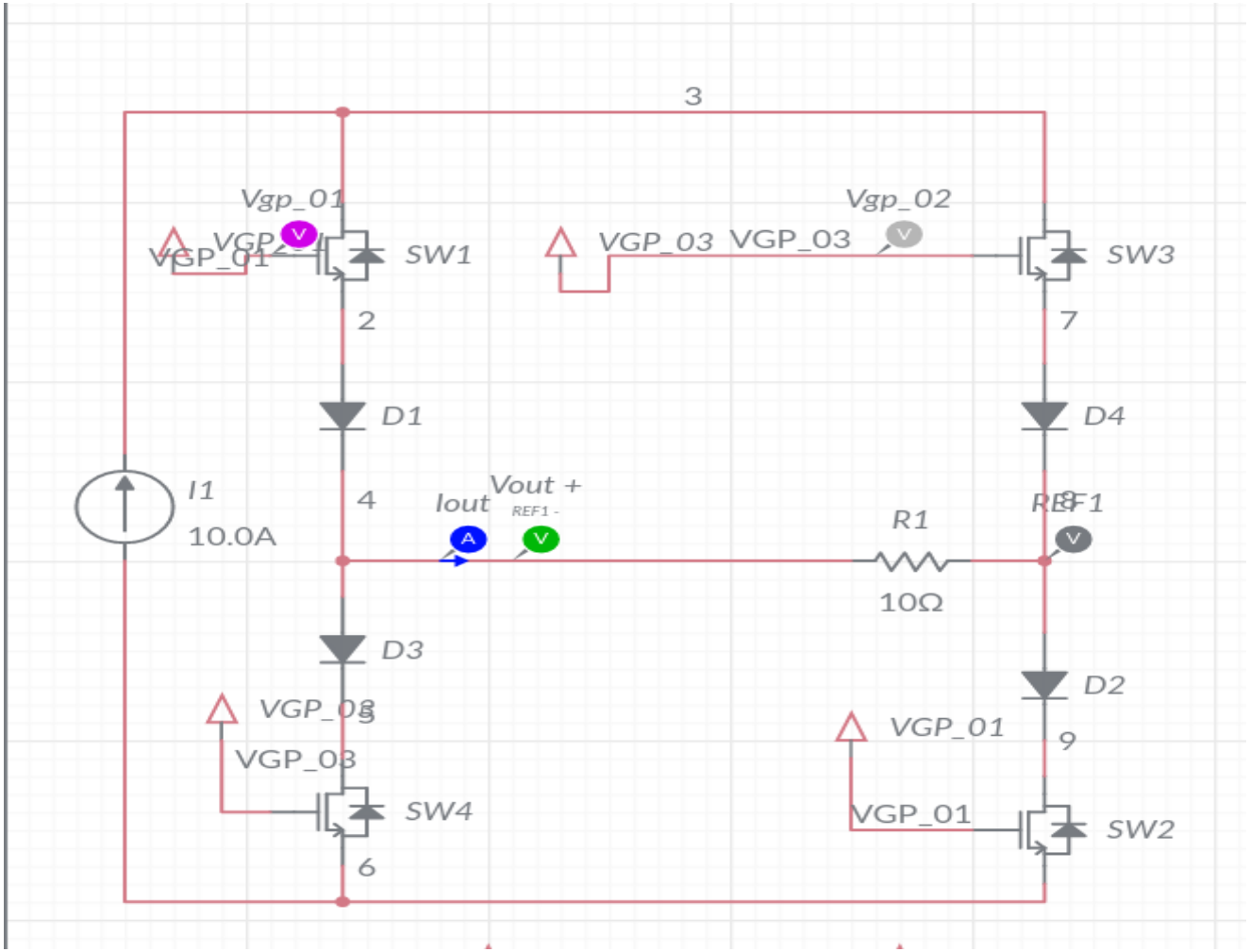


Figure: VSI inverter circuit

recap. VSI

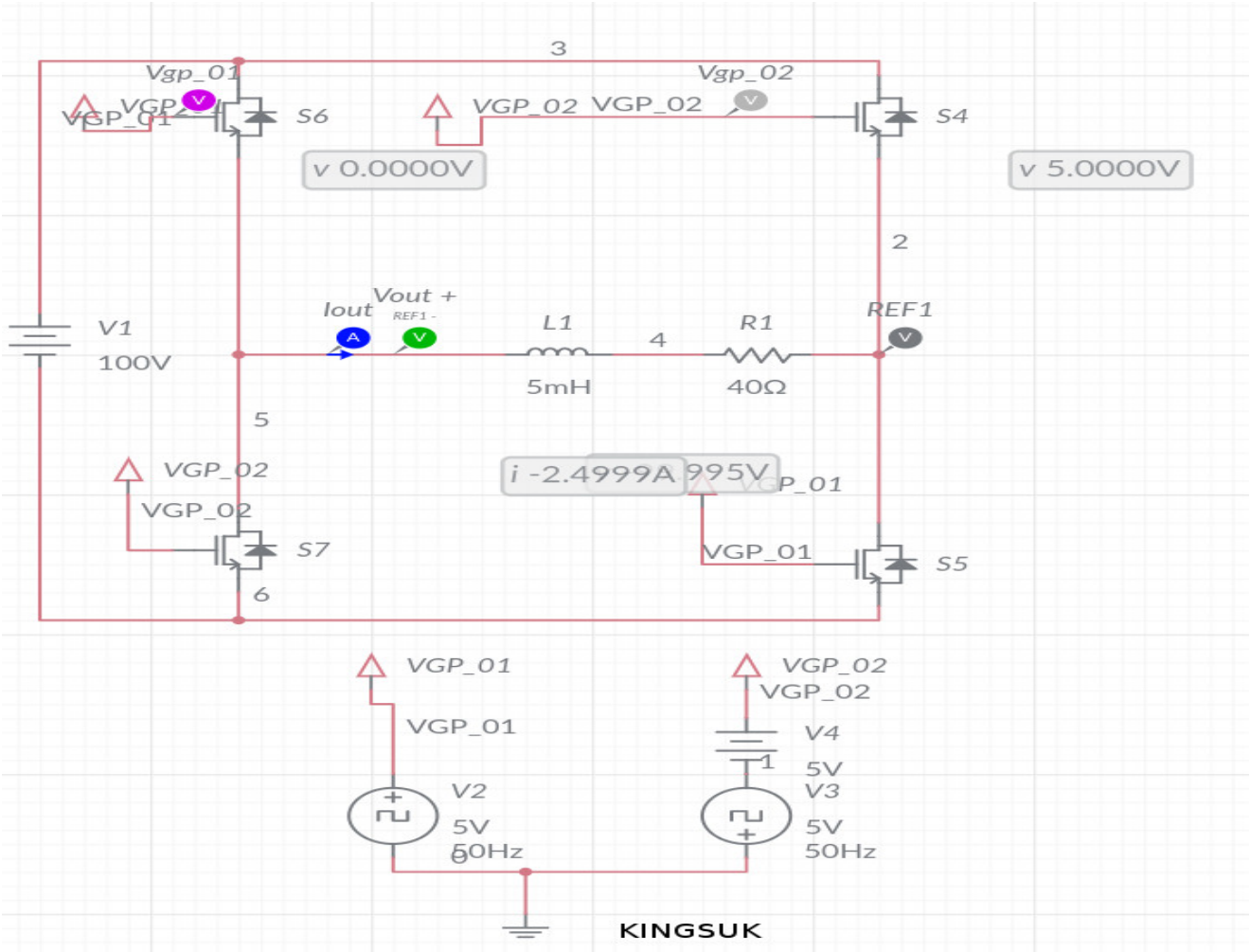


Figure: CSI inverter circuit

Wave form of CSI

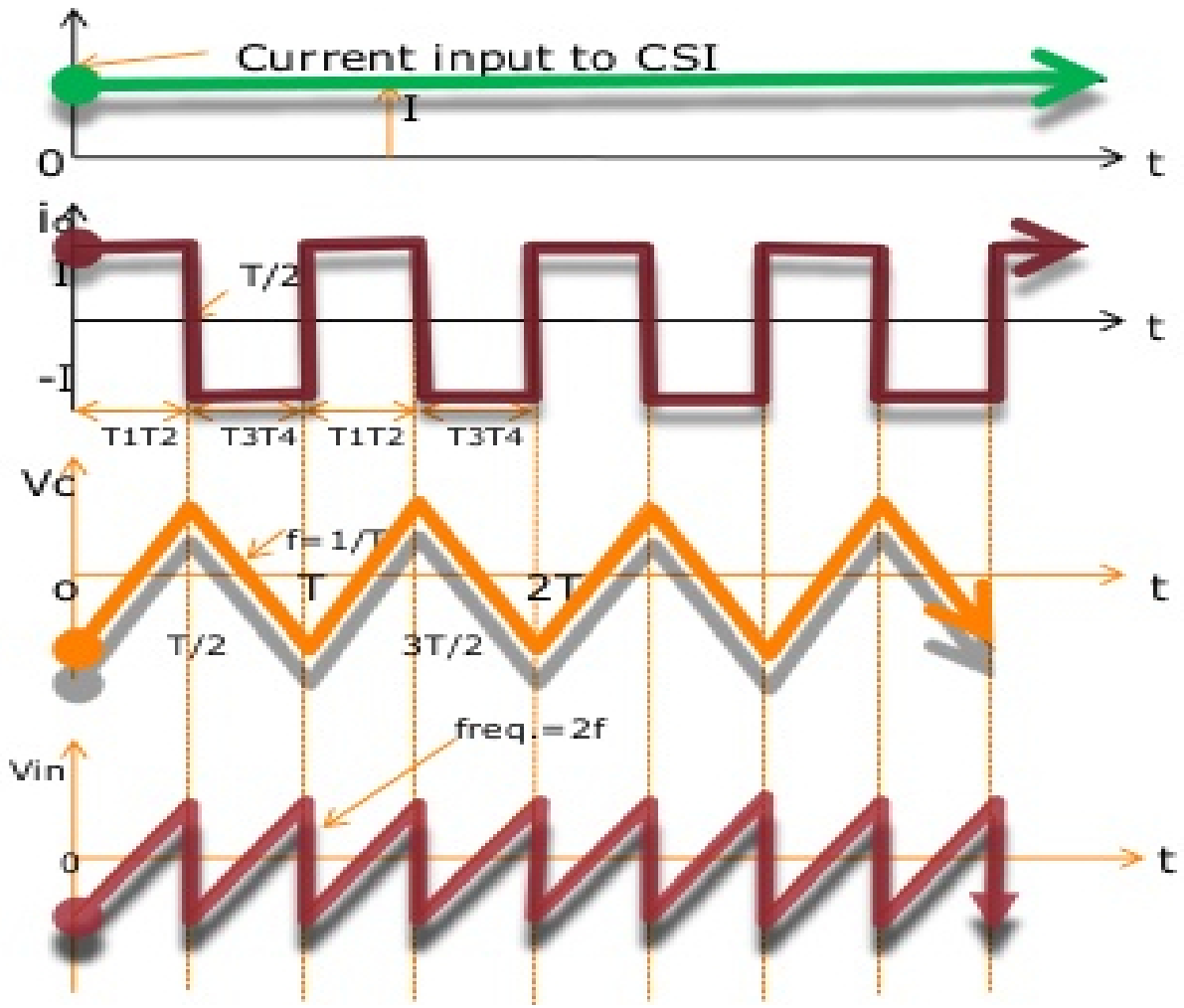


Figure: CSI-wave

NB:picture taken from google

Load current in CSI

$$i_0 = \sum_{n=1,3,5..}^{\inf} \frac{4I_L}{n\pi} \sin(\frac{n\delta}{2}) \sin(n\omega t) \tag{1}$$

Advantages of CSI

- As input DC current is controlled and limited hence misfiring of the switches, short circuit are NOT a serious problems.
- The peak current of power device is limited.
- The commutation circuit of the switch devices (say Thyristors) are simpler.
- It has the ability to handle reactive or regenerative load without free wheeling diodes.

References


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